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A Schema-Driven Prompt Engineering Approach for Data-Centric LLM Tasks

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***Abstract*—General-purpose LLMs are increasingly employed in a variety of data-centric tasks, ranging from entity and relationship extraction from plain text to schema-aware data integration and analysis. Their large number of parameters and extensive training corpora enable strong generalization across application domains. However, LLMs lack access to up-to-date knowledge and are prone to hallucinations, limiting their reliability in data management scenarios. To address these issues, prompt engineering techniques have been proposed to specify the context in which a task should be performed. In this paper, we explore the use of conceptual schemas as a foundation for schema-driven prompt engineering, providing structured and reusable contexts for grounding LLM behavior in data-centric applications. We present SchemaLink, an intelligent web-based system for the graphical design and enhancement of conceptual schemas and their exploitation for LLM-based tasks. We demonstrate the applicability of our approach to knowledge extraction from plain text and to the discovery of joinable columns in data lakes, showing how schema-driven prompting improves grounding and consistency across heterogeneous data sources.**

***Index Terms*—LLM contexts, schema curation, named entity extraction, relation extraction, and joinable column discovery**

I. Introduction

LLMs [1] are advanced neural networks that have attracted significant attention due to their strong performance across a wide range of natural language processing (NLP) tasks. Their capabilities in language understanding and content generation arise from training models with billions of parameters on massive corpora of textual data, enabling them to capture linguistic patterns, grammar, contextual dependencies, and a shallow form of reasoning. Several general-purpose LLMs (e.g. OpenAI's GPT-4, Google Gemini, Meta's LLaMA, and Mistral) are currently available through both commercial providers and open-source communities. These models differ in terms of architecture, scale, and performance, reflecting variations in design choices, training data, and parameter counts.

Despite their impressive capabilities, LLMs suffer from two major limitations: the lack of access to external and up-to-date knowledge, and the tendency to produce hallucinations. The former limitation is due to static knowledge acquired at training time and therefore cannot natively incorporate information generated afterward. Hallucinations occur when a model produces outputs that are syntactically well-formed but factually incorrect, inconsistent, or entirely fabricated. This behavior arises because LLMs generate responses by modeling statistical regularities in their training data, rather than by grounding outputs in verified external knowledge or a true understanding of the domain.

Prompt engineering [2] has emerged as a primary strategy for mitigating these limitations by systematically designing instructions and contextual information that guide LLM behavior. A variety of techniques have been proposed, including: zero-shot, one-shot, and few-shot prompting [3], where models are guided with none or only a small number of examples; retrieval-augmented generation (RAG) [4], which integrates information retrieval mechanisms to extend the model's effective knowledge beyond training-time data; and more advanced strategies such as chain-of-thought [5], tree-of-thought [6], and self-consistency prompting [7]. While effective, these techniques are largely task-specific and often rely on manual prompt design, motivating the need for structured, reusable, and systematic prompt engineering approaches.

Conceptual schemas [8] have long been adopted in the database community to represent domain entities and the relationships among them. In this work, we argue that conceptual schemas can serve as a principled foundation for schema-driven prompt engineering, enabling the explicit specification of the structural and semantic context in which an LLM operates. A schema captures the structural aspects of relevant entities and relationships and can be enriched with metadata, such as example instances and references to established ontologies, to better ground real-world concepts in standardized classes and properties. Once defined, a conceptual schema can support multiple downstream applications that exploit LLM reasoning capabilities. Specifically, schemas can be used to systematically generate prompts for a class of data-centric LLM tasks, in which model outputs must conform to structured data representations, schemas, or query languages. Examples include entity and relationship extraction from plain text, semantic annotation of tabular data, and the translation of natural language queries into SQL, SPARQL, or Cypher expressions. By providing an explicit and reusable context, schemas reduce ambiguity, improve grounding, and increase the consistency of LLM outputs across tasks.

In this paper, we present a schema-driven prompt engineering approach and the main research efforts conducted at the

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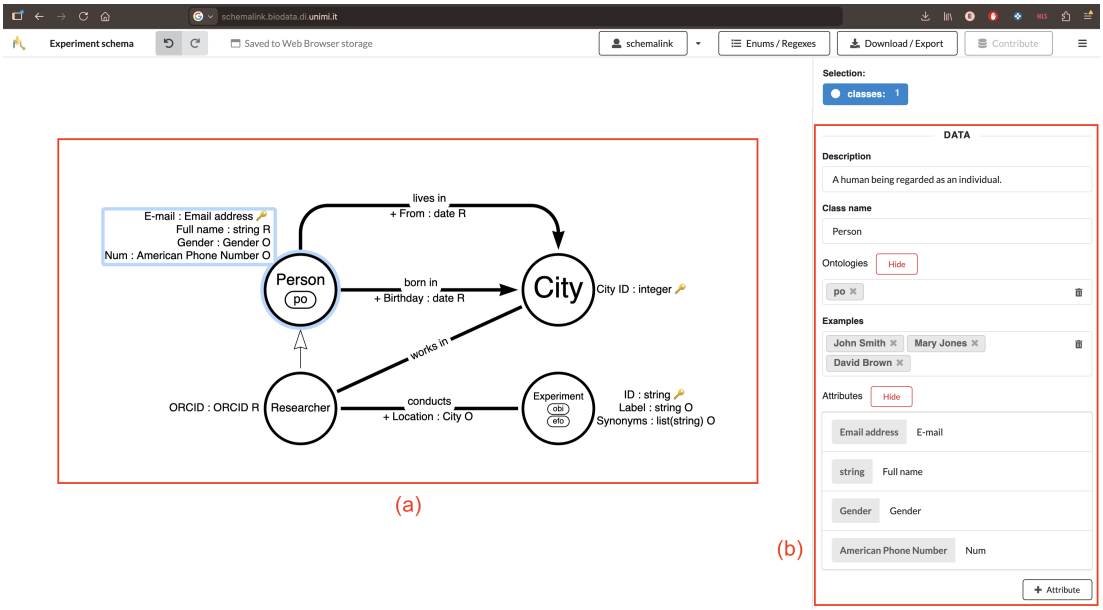


Fig. 1: Graphical representation of a conceptual schema (a) and sidebar for the specification of class properties (b).

University of Milan to realize this vision. SchemaLink [9], a web-based environment for the graphical construction and enhancement of conceptual schemas, is introduced in Section II. We adopt LinkML [10] as the underlying schema representation formalism, as it provides a YAML-based language for describing domain entities and relationships, along with example instances and ontology-based annotations. The textual nature of LinkML further facilitates the systematic generation of LLM prompts from schema components. Section III shows the use of conceptual schemas for entity and relationship extraction from plain text, reporting promising results in the biomedical domain [11], [12]. Finally, Section IV demonstrates how conceptual schemas can be leveraged for the semantic annotation of tabular datasets and for discovering joinable columns in data lakes using LLMs [13]. Final remarks are reported in Section V.

II. SchemaLink

SchemaLink is a web-based environment for the graphical construction and enhancement of conceptual schemas (`SchemaLink.biodata.di.unimi.it`). SchemaLink extends the `arrows.app` [14] graphical environment proposed by Neo4j with dedicated functionalities for the specification of classes, associative and inheritance relationships, and schema constraints (e.g., primary keys, mandatory and optional properties, cardinality constraints, descriptions and examples taken from ontologies or knowledge graphs). The graphical language resembles the one adopted in UML/ER-like models for the specification of classes and relationships, but leverages the simplicity of `arrows.app` for sketching nodes and relationships among them.

Example 1 *Suppose a user wishes to define a schema describing researchers, the experiments they conduct, and the cities they are associated with. Fig. 1.a illustrates a schema that includes the classes* `Person`, `City`, `Experiment`, *and* `Researcher`, *where* `Researcher` *is defined as a* subclass of `Person`. *The associative relationships named* `born in`, `lives in`, `conducts`, *and* `works in` *are defined among classes. Classes and relationships may have attributes (e.g., the full name of a person, the date from which a person lives in a given city) as well as cardinality constraints. Moreover, SchemaLink provides a sidebar that enables the specification of properties for classes and relationships. For example, when the class* `Person` *is selected in the canvas (Fig. 1.b), the sidebar displays its description, references to ontologies, example instances, and set of attributes.*

SchemaLink integrates a RAG-based approach [9] to assist in creating new schemas from scratch and enhancing existing ones. A collection of LinkML schemas are indexed using the OpenAI's *text-embedding-3-large* model and used as *context* for providing useful suggestions to a general-purpose LLM. In our experiments, we have considered 100 LinkML schemas developed in Monarch projects [15] and adopted the *gpt-4o-mini* model because it offers a good trade-off between output quality and latency times.

For refining a schema S, different kinds of schema modification query templates have been formulated (like adding a class, including new relations, reifying an attribute, explaining the role of a class/relation/attribute in the schema) that can be issued to *gpt-4o-mini* along with schema samples similar to S to suggest a new modified version. Depending on the portion of the schema selected by the user, a specific schema mod-

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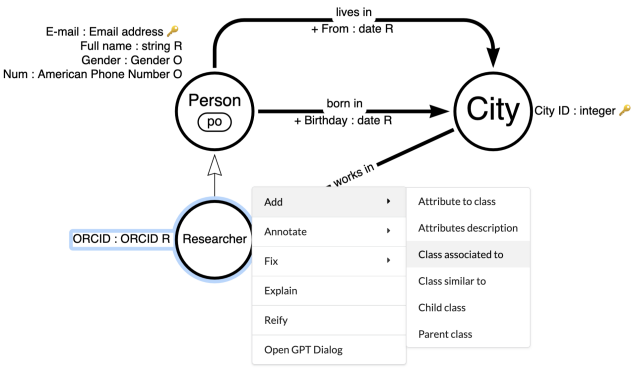
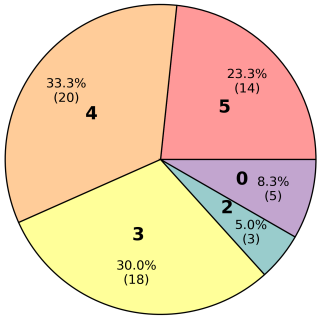


Fig. 2: Intermediate schema generated from SchemaLink and menu with intelligent operations.



	Disease	Drug	Protein	Pathway	RNA
DeepSeek	3.7	4.5	3.5	4.0	3.3
Gemini	3.3	3.7	3.2	2.3	3.5

Fig. 3: Average scores for intelligent editing operations.

ification query template is selected, instantiated, and issued to the LLM along with schemas that are similar to S. After the validation of the proposed schema (to avoid hallucination issues), it is shown to the user for the final approval.

Example 2 *Suppose the user wishes to generate the schema shown in Fig. 1 from scratch. Using the SchemaLink intelligent component, the user can issue the request:* "Generate a schema involving researchers, modeled as a specialization of a class Person, and the cities they work in. Moreover, include plausible relationships between Person and cities." *that produces the the intermediate schema in Fig. 2. Then, the user can further refine the intermediate schema through the set of predefined operations. By right-clicking on the class* `Researcher`*, the user selects the operation* Add class associated to*, which introduces the class* `Experiment` *and the corresponding* conducts *relationship with* `Researcher`*.*

Although similar approaches for graphical schema curation have been proposed in the literature [16]–[18], to the best of our knowledge, none of them incorporate artificial intelligence features to assist end-users in designing schemas.

A key issue in the use of LLMs for proposing modifications to a schema is the evaluation of the quality of the generated schemas. Relying on domain experts is often unfeasible in real-world settings, due to both the limited availability of qualified experts and the costs associated with their involvement. For this reason, we have envisioned the use of LLM-as-a-Judge [19] approach for evaluating the quality of the obtained schemas. In a biomedical project, we defined five use cases for the creation of conceptual schemas and prompted SchemaLink to generate an initial schema. The use-case models concern: (i) entities involved in diseases (`Disease`); (ii) drugs and their mechanisms of action (`Drug`); (iii) proteins and their functions (`Protein`); (iv) biological pathways and their participating entities (genes, proteins, metabolites; `Pathway`); and (v) relations among RNA molecules (`RNA`). Then, we prompted 2 LLMs (DeepSeek and Gemini) to act as expert biomedical curators and assign a discrete score on a five-point scale: 1 = schema unintelligible or not applicable to the domain; 2 = largely incorrect with major inconsistencies; 3 = broadly usable but requiring several revisions; 4 = correct and coherent with only minor refinements needed; 5 = fully correct and well-suited to the domain. Using independent LLM judges improves fairness and reduces vendor-specific bias (e.g., avoiding the situation in which only ChatGPT is used to evaluate a schema produced by a GPT-based engine). Table I reports the scores assigned by LLM-judges across the five use cases. Overall, scores are consistent across models: all ratings are greater than or equal to 3, and in three use cases a score of 4 or higher was assigned by at least one judge.

Starting from the schemas generated in the previous step, LLM judges were then iteratively prompted to select and evaluate six intelligent editing operations. Resulting schemas were scored on a 0–5 scale. Scores from 1 to 5 follow guidelines analogous to those used for the initial schema evaluation, 0 indicates a failure to produce any visible or meaningful schema modification. Fig. 3 summarizes the average scores obtained for each case and judge. Out of 60 evaluations, 52 (86.6%) achieve a score greater than or equal to 3. Overall, these results show that the schema refinements produced through intelligent editing operations are effective for the targeted schema curation tasks. A manual inspection of high-scoring interactions further confirmed that the proposed refinements are reasonable and align well with existing domain schemas, including those curated and stored in the RAG database. We remark that the judges assigned a score of zero in five evaluations, all of which concerned operations to modify attribute types. In these cases, the target attributes already had appropriate types, and no modification was thus needed. After re-prompting the LLM judges to consider this condition explicitly, they clarified that the low scores were attributable to an inappropriate choice of operation rather than to a failure of the intelligent editing component.

	Disease	Drug	Protein	Pathway	RNA
DeepSeek	3	4	3	3	3
Gemini	4	4	3	4	3

TABLE I: Scores for schemas generated via SchemaLink.

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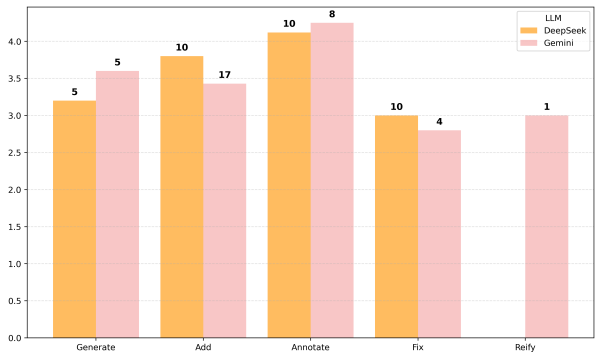


Fig. 4: Scores of intelligent operations by category. Numbers above each bar indicate how many times they were invoked.

Fig. 4 reports the aggregated scores assigned by the LLM judges to the intelligent operations, grouped by operation category. The number displayed on top of each bar indicates how many times the corresponding operation was selected and evaluated. The results show that *Annotate* and *Add* operations achieve higher scores than other categories. An explanation is that these operations benefit from a larger and better curated set of relevant examples stored in the RAG, and are invoked more frequently by the models. This is consistent with preliminary feedback collected from human curators, who reported that these operations are the most commonly requested.

III. NER and RE from Plain Text

Named Entity Recognition (NER) and Relation Extraction (RE) are core NLP tasks [20], [21] that underpin the construction and enrichment of Knowledge Graphs (KGs). Accurate identification of entities and their semantic relations enables the transformation of unstructured text into structured, machine-interpretable knowledge.

In [12], [22], we proposed a schema-driven prompt engineering method for NER and RE from plain text to populate a biomedical KG tailored for representing RNA interactions [23], [24]. The conceptual schema describes the kinds of entities (classes) and relationships that we wish to extract from textual documents and is converted into a list of prompts that guide LLM outputs toward a predefined structure, enabling zero-shot or few-shot extraction. The approach improves structural coherence, facilitates validation, and reduces interpretation errors, making extraction more reliable and reproducible. The approach is articulated in two phases.

In the first phase, the list of prompts summarizes the relevant information useful for the extraction of the corresponding entities and relationships. Moreover, they contain examples of instances and descriptions of associated ontological classes for grounding the identified entity mentions. Dependencies among prompts (like the prompt for identifying relationships between two entities should be issued after the prompts for identifying the involved entities) are taken into account to optimize the retrieval approach. Prompts are also parametrized to consider the information previously extracted that might influence the LLM extraction process. After the invocation of each prompt, each mention is linked to a concept of the corresponding ontology or identification scheme. This activity is particularly relevant for obtaining a homogeneous representation of the facts that should be included in the target KG.

Example 3 *Suppose the user is interested in extracting facts according to the schema in Fig. 1. Fig. 5.a shows the prompt generated for class* `Experiment` *(analogous prompts can be obtained for the other classes). The prompt for* `Experiment` *is enriched with representative examples retrieved from the Experimental Factor Ontology (EFO [25]) and the Ontology for Biomedical Investigations (OBI [26]), which annotate* `Experiment` *in the schema. By contrast, Fig. 5.b shows an example of prompt generated for the relationship* `conducts`*. This prompt contains some parameters that will be filled at execution time, when the instances of the involved classes will be determined. This prompt explicitly constrains the LLM to consider only the lists of source and target entity mentions identified in the previous steps.*

The prompts will be generated independently from the texts from which entities and relationships will be extracted.

Example 4 *Suppose now that the generated prompts should be evaluated on the following text:*

> "Dr. John Smith (e-mail: john.smith@unimi.it; ORCID: 0000-0000-0000-0111) conducted an in vitro CRISPR screen experiment, investigating the role of TP53 in cancer progression. The experiment followed gRNA-seq and other targeted gene-editing assays previously reported in the literature."

In this case, no instances of class `City` *can be identified and therefore the prompts associated with the relationships involving a city can be skipped. The invocation of the prompt of the class* `Person` *allows the identification of the mention John Smith, mapped to:* `john.smith@unimi.it` *that is then specialized as an instance of class* `Researcher`*. The invocation of the prompt of the class* `Experiment` *permits the extraction of the mentions in vitro CRISPR screen experiment and gRNA-seq. Mentions of type* `Experiment` *are then grounded via label–identifier lookup tables derived from the ontologies specified in the schema; accordingly, these mentions are mapped to* `OBI:0003659` *and* `EFO:0030033`*. Notably, the grounding process mitigates ambiguities in the original text (e.g., targeted gene-editing assays is a broad concept absent in the adopted ontologies), while limiting LLM hallucinations.*

Before invoking the prompt of the relationship `conducts`*, the parameters are instantiated (red slots in Fig. 5.b). The result is: John Smith-conducts-in vitro CRISPR experiment, mapped to:* `john.smith@unimi.it-conducts-OBI:0003659`*; John Smith-conducts-gRNA-seq, mapped to:* `john.smith@unimi.it-conducts-EFO:0030033`

The adoption of schema identifiers and ontologies makes the extracted triples directly suitable for insertion into KGs that adhere to them.

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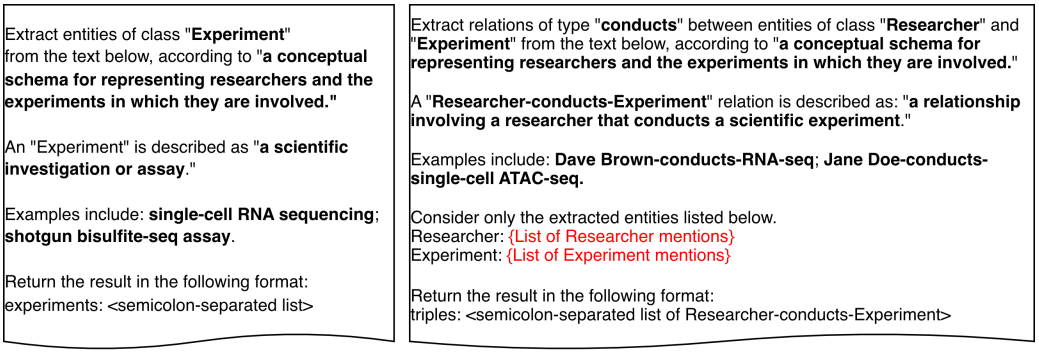


Fig. 5: Examples of prompt for extracting entity mentions (a) and relationships (b).

Despite schema- and ontology-driven guidance, LLMs may still generate facts that violate domain constraints or introduce novel but plausible facts. To filter and validate these outputs, we apply graph machine learning techniques (GRL) [27] to assess their plausibility with respect to a reference KG. Unlike traditional evaluation metrics that treat such cases as false positives, our approach recognizes that some of these "violations" may reflect correct but unannotated knowledge, enriching the KG beyond the reference annotations.

Example 5 *To validate the triples generated in Example 4, a KG constructed from scholarly sources (e.g., Semantic Scholar, Google Scholar, ResearchGate, and PubMed) can be exploited, where nodes include researchers and experiments, and edges include* `conducts` *relationships observed in the literature.*

Given the grounded triples for the `conducts` *relationships identified in the previous example, the KG can be queried to identify the presence of identical or semantically equivalent edges. Whenever they are missing, a link prediction model trained on the KG can be exploited to estimate the* plausibility *scores of each candidate triple. These scores enable ranking the extracted triples: high-confidence candidates can be prioritized for automatic insertion or light-weight validation, medium-confidence ones can be flagged for curator review, and low-confidence ones can be discarded.*

The approach has been tested on a dataset of 100 manually annotated documents [28]. Using GPT4-turbo as back-end LLM and DeepWalk [29] as KG embedding model we reached F1-scores above 90% in RE task for specific relations. We also compared the approach with base LLMs. Results reported in [12] show that our method consistently outperforms gpt-3.5-turbo and LLaMA 2–70B-chat used in isolation, achieving improvements exceeding 20% in F1-score for RE. Finally, combined with DeepWalk [29] for KG embedding, the approach is able to effectively distinguish high-, medium-, and low-confidence candidate relations [12]. This means that expert curators can check with attention only medium-confidence candidates, while the others can be directly discarded and accepted.

IV. Joinable Table Discovery

Data lakes [30] have recently emerged as critical infrastructures for storing and managing large, diverse, and rapidly growing collections of data. Unlike traditional data warehouses, which require data to conform to a predefined schema before ingestion, data lakes store data in its native format with little or no upfront modeling. While this flexibility enables scalable and cost-effective data integration, it also introduces significant challenges for data discovery and integration tasks. In particular, identifying joinable columns across tables becomes difficult due to the high heterogeneity of tabular datasets, inconsistent naming conventions, and the absence of standardized schemas or shared ontologies [31].

Traditional join operations that rely on exact matches are often ineffective in data lake environments, semantically equivalent columns may differ substantially in their names, data formats, or value representations. Such variability prevents conventional syntactic matching techniques from correctly identifying meaningful join relationships. To address this limitation, several approaches (e.g., Pexeso [32], DeepJoin [33], WarpGate [34]) have been proposed that leverage neural models to learn representations of datasets in a latent space. These methods encode columns as vectors, where semantic similarity is captured through geometric proximity. Columns whose vector representations are close to one another are therefore more likely to share similar semantics and can be identified as candidates for join operations.

Although these methods can be reliable when the collection of tabular datasets is homogeneous and organized in the same contextual domain [35], they still struggle to deal with heterogeneous collections because they do not consider the context and the metadata associated with every single dataset. To address this challenge, in [13] we proposed the use of conceptual schemas to explicitly model the entities and relationships relevant to a specific type of analysis. These schemas serve as a semantic reference framework and are used to annotate the datasets stored within a data lake, thereby enabling more effective analytical tasks. A three-step pipeline has been then realized.

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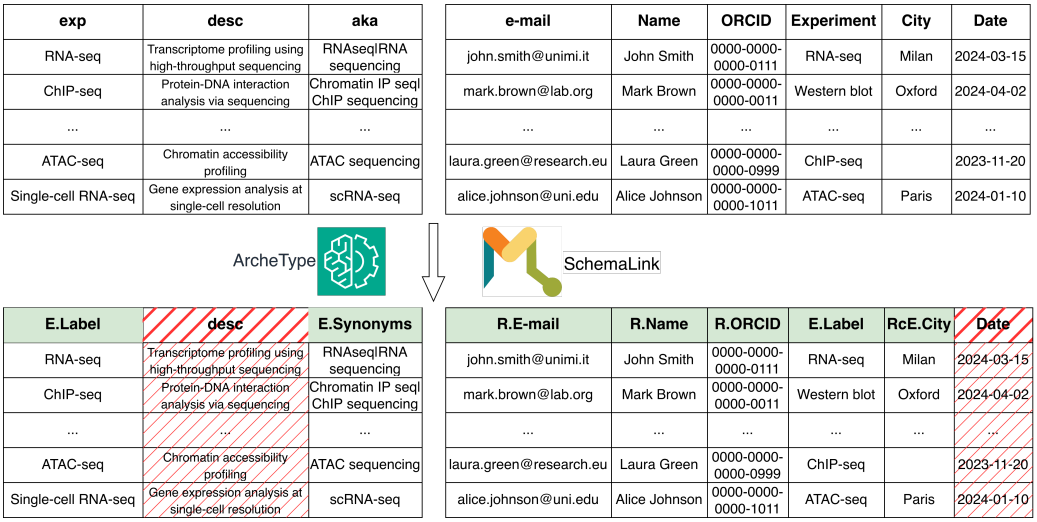


Fig. 6: Example of semantic annotation process.

The first step is devoted to the semantic annotation of relevant datasets by adapting ArcheType [36] to work with our conceptual schemas and annotate dataset columns. ArcheType exploits LLMs applied on samples of values extracted from the different columns for the annotation. This procedure offers a twofold advantage. First, columns are semantically labeled using the class properties defined in the conceptual schema. Second, datasets and columns that are irrelevant to the target analysis can be automatically identified and pruned, reducing noise and improving efficiency. In addition, LLMs are used to enrich and improve dataset metadata that, in real scenarios, are often incomplete, inconsistent, or poorly structured.

Example 6 *Fig. 6 shows two tabular datasets on which the first step of our procedure is applied by considering the conceptual schema in Fig. 1. Columns are annotated with schema attributes using ArcheType. For example, columns describing experiment labels and synonyms are mapped to attributes of the class* `Experiment` *(e.g.,* `E.Label`*); researcher e-mail addresses and ORCID identifiers are mapped to attributes of the class* `Researcher` *(e.g.,* `R.ORCID`*); and the cities where experiments were conducted are mapped to the inlined attribute* `City` *of the* `Researcher-conducts-Experiment` *relationship (i.e.,* `RcE.City`*). Moreover, the columns* `desc` *and* `Date` *are not explicitly grounded in the schema and are therefore pruned.*

After semantic column annotation, the pipeline proceeds with the column embedding stage. In this phase, each column is mapped to a vector representation in a high-dimensional latent space using a representation model that integrates multiple sources of contextual information. Specifically, the embeddings are computed by leveraging the enriched metadata, the conceptual schema, and the semantic annotations associated with each column. This unified representation enables the computation of similarity measures (e.g., cosine similarity) between column embeddings, which can be exploited to identify candidate join relationships. Consequently, the resulting embeddings encode not only surface-level syntactic features, but also deeper semantic properties and structural relationships within the overall data lake context.

In the last phase, a *semantic catalog* is constructed as a property graph whose nodes correspond to the annotated datasets, and their properties represent metadata associated with the datasets. The graph edges identify the joinability conditions. The columns that can be used for the join operation are determined relying on the proximity of their embeddings in the latent space. This graph can be utilized for conducting all the analyses involving the entities and relationships represented in the associated conceptual schema.

Although defining a schema may initially seem burdensome, particularly within the schema-less paradigm of data lakes, we argue that schemas play a crucial role in narrowing the search space to datasets that are most relevant for specific analytical tasks. Moreover, schema development is simplified through our SchemaLink editor. Another common barrier to schema adoption is the cost associated with semantic annotation. However, ArcheType has demonstrated that this process can be performed efficiently when datasets are acquired and, when combined with appropriate indexing structures, can substantially improve results for the downstream join detection task.

To evaluate the impact of semantic annotations in join discovery, we considered three benchmark data lakes (WebTables [37], WikiTables [38], and ECB [39]) that represent complementary scenarios. For each of them, Table II reports the number of tables, average number of columns per table, average number of rows, and average number of annotated joins. The first two benchmarks consist of relatively small and "short" tables, both in terms of number of columns and rows, and provide curated ground truth annotations for joinability. For these datasets, we considered tables involved in at least

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TABLE II: Descriptive Statistics of Data lakes.

Dataset	Tables	Avg. Columns	Avg. Rows	Avg. Joins
WebTables	$4.3k$	12.9	64	12.7
WikiTables	$3.5k$	2.8	35	0.9
ECB	74	34.0	$1.7M$	927

one join to reduce the bias induced by the evaluation of trivial non-join cases. Despite their similar scale, the average number of joins in WebTables is high (almost every column within a table is joinable with a column in another table); conversely, WikiTables exhibits a much sparser join structure, with only a small fraction of columns (roughly one out of three) being joinable. By contrast, ECB consists of a small number of large, wide tables with millions of rows and a high number of columns. Pairs of columns with identical headers are considered join candidates.

Even if these benchmarks are usually adopted in joinable table discovery, their use in our data lake context with heterogeneous headers and values presents important limitations. Specifically, join annotations often rely solely on header equality, which does not always correspond to true semantic joins in real-world scenarios. Identical headers may refer to different concepts depending on the context (e.g., a column named *id* might denote a personal identifier, a surrogate key, or a tax code). Moreover, many tables in WebTables and WikiTables are generated by horizontally splitting datasets into multiple fragments to induce join relationships. While this process increases the joinable column pairs, it also amplifies header reuse across tables. For example, the attribute *GIDP* (Grounded Into Double Play) appears as a valid join key across multiple baseball statistics tables. Although columns with this header are joinable in this specific context, the same concept may be expressed using lexical variants (e.g. *GIDoubPl*, *GroundedIntoDoublePlay*). As a consequence, join discovery models may implicitly learn to associate columns based on header equality rather than true semantic compatibility.

For each dataset, we split columns into training and test sets. We adopt different train/test ratios depending on dataset size and join density. Specifically, for WebTables we use a 90/10 split, for WikiTables, we adopt a 60/40 split, for ECB, we use an 80/20 split. Then, training sets are semantically annotated via Archetype (we considered *gpt-oss:20b* as backend model because it is open-source, free to use, and efficient). Then, to overcome the presented limitations and to assess the impact of semantic annotations in realistic data lake scenarios, we introduce the following controlled corruptions in the test sets: replacing column names with generic placeholders (e.g., *Column_0*, *Attribute_1*, *Var_X*); injection of character-level noise into column names (i.e., swapping adjacent characters, randomly deleting one character, replacing a character with a nearby one on a keyboard); shuffling columns' headers. This setting simulates a common scenario in which data lakes can be curated offline and used to train a ML model, whereas newly ingested tables arrive with noisy or unreliable headers.

Fig. 7 reports the average $F_1@5$ and $F_1@10$ scores obtained

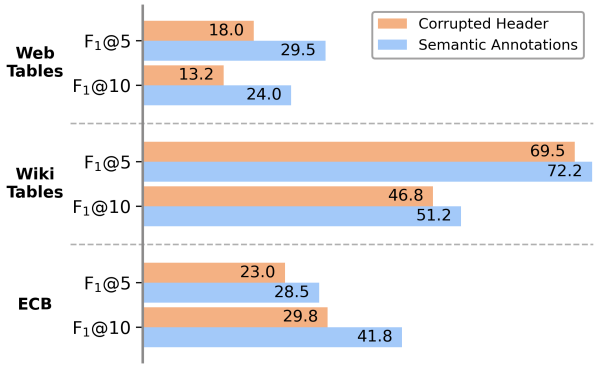


Fig. 7: DeepJoin F_1-scores on corrupted test sets' headers with semantic annotation applied at training time.

using "corrupted" headers and semantic annotations when applying DeepJoin. Across all datasets, semantic annotation consistently improves performance. Importantly, the observed gains with DeepJoin are achieved even though only the training set is semantically annotated. This result highlights a key practical advantage of our approach: semantic annotation does not need to be applied to newly ingested tables. Instead, annotation (which is a time-consuming activity) can be performed once during training.

V. Conclusion and Future Work

In this paper, we presented a schema-driven prompt engineering approach for supporting data-centric tasks with LLMs. Our work builds on the observation that, while LLMs exhibit strong reasoning and generation capabilities, their effectiveness in data management scenarios critically depends on the availability of explicit structural and semantic context. We argued that conceptual schemas provide a principled and reusable mechanism for specifying such context, enabling more grounded, consistent, and controllable LLM behavior across heterogeneous data-centric applications.

We described a set of complementary contributions that operationalize this vision. First, we introduced SchemaLink, a web-based environment that facilitates the design, refinement, and reuse of conceptual schemas using the LinkML formalism. By combining graphical interfaces with AI-assisted support, SchemaLink lowers the barrier to schema creation and promotes the adoption of schema-driven approaches. Second, we demonstrated how conceptual schemas can be systematically exploited to guide prompt construction for entity and relationship extraction from plain text, achieving promising results in the biomedical domain. Finally, we showed how this schema-based framework can be used to the semantic annotation of datasets and to the joinable columns discovery in data lakes.

Overall, our results indicate that schema-driven prompt engineering effectively improves grounding, reduces ambiguity, and enhances the quality of LLM outputs in data-centric tasks. By explicitly encoding domain structure and semantics, schemas enable LLMs to better align generated outputs with

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the intended data models, thereby mitigating hallucinations and increasing interoperability.

Several directions for future work remain open. From a methodological perspective, we plan to further automate the generation and refinement of prompts from schemas, as well as to introduce further logical constraints that can be checked through the generated prompts. From a systems standpoint, we aim to study scalability aspects in large data lake settings and to integrate more advanced indexing and retrieval mechanisms. Finally, we intend to explore tighter integration with query processing and data integration pipelines. We believe that schema-driven prompt engineering represents a promising step toward principled, reliable, and reusable use of LLMs in data management.

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